

Ergonomic and Flexible Paperclip Design

Abstract

This paper explores a novel design concept for a paper clip, combining functionality and ergonomic efficiency. The design incorporates a central metal strip/metal base to provide structural integrity, complemented by a twistable/bendable metal with rubber casing, enabling enhanced flexibility and grip. The inclusion of an empty space in the core of the clip facilitates easy handling and efficient attachment of documents. This innovative approach addresses common limitations of traditional paper clips, offering improved usability, durability, and adaptability. The proposed design aligns with sustainable practices, leveraging durable materials while maintaining a lightweight structure. This study highlights the potential of modern materials and thoughtful engineering to elevate everyday office tools.

Technical Field of Invention

Office Supplies and Tools. More specifically, it pertains to Ergonomics, Product Design, and Materials Engineering. It

involves aspects of mechanical design, material science, and user-centered design for office accessories and tools. The invention may also touch on sustainability and durability in everyday office products.

Cooperative patent classification

A44B 19/20 - Fastening devices for papers: This subclass is related to clips or similar devices for fastening papers. Your design, which includes a new take on the paper clip, would likely fall under this category.

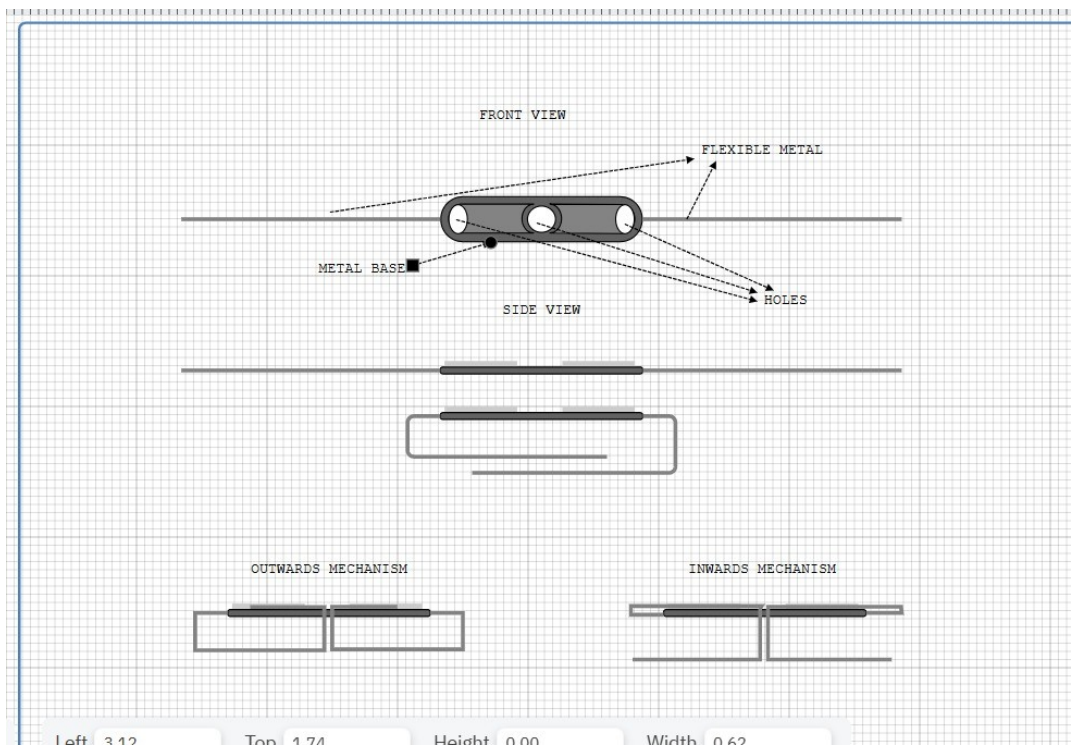
A44B 19/26 - Paper clips with special features: This classification specifically refers to paper clips that include unique or innovative designs, such as the twistable/bendable feature of your design.

B65D 75/00 - Articles of office or stationery equipment: This class involves various office tools, including paper clips, with a focus on their structural design and functionality.

A47B 97/02 - Ergonomic office tools: If your design emphasizes the ergonomic aspect, such as ease of use and handling, it may be relevant for ergonomic-focused classifications.

F16B 2/00 - Clamps or other devices for holding objects together: Though this is a broader classification, it may also apply to your paper clip design because of its mechanical aspect of holding objects (documents) together.

C21D 1/00 - Metalworking – Properties or treatment of metal: If your design includes specialized metalworking or unique metal treatments (such as flexible metal with rubber casing), this might apply.



Prior art search

1. <https://utppublishing.com/doi/pdf/10.3138/guthrie.64.4.146>
2. <https://www.patentrenewal.com/post/nordic-inventions-patents-for-the-paper-clip>
3. <https://edworking.com/news/startups/paper-clip-history-from-early-patents-to-modern-innovations>

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